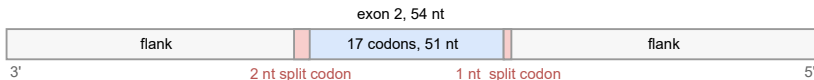


A The library: BRCA1 exon 2, three mutation types, P5/P7 adaptors

chr17:43,115,634-43,115,878 minus strand, 245 nt

17 codons $\times$ (19 aa + 1 in-frame deletion + 1 stop) = 357 oligos

P5 20 nt + 245 nt + P7 21 nt = 286 nt (283 where a codon is deleted)

B PoolParty

```
import poolparty as pp

ADAPTOR_5 = "AATGATACGGCGACCAACCGA"
ADAPTOR_3 = "TCGTATGCCGCTCTTCGTCTTG"

# Extract region and annotate the target CDS
orf = pp.from_fasta(
    "chr17.fa", (43115633, 43115878, "-")
).annotate_orf(region_name="cds", extent=(100, 151), frame=1)

# Amino-acid substitutions
aa = orf.mutagenize_orf(
    region="cds", num_mutations=1,
    mutation_type="missense_only_first", mode="sequential")

# In-frame codon deletion scan
inframe = orf.deletion_scan_orf(
    deletion_codons=1, deletion_marker=None,
    region="cds", mode="sequential")

# Stop codon scan
stop = orf.insertion_scan_orf(
    "TGA", region="cds", replace=True, mode="sequential")

# Stack three pools, then add the adaptors
library = pp.join([ADAPTOR_5, pp.stack([aa, inframe, stop]),
                  ADAPTOR_3])

library_df = library.generate_library()

Inputs: 1 FASTA
Outputs: oligos, with optional design card parameters that
produced them
```

C VaLiAnT

```
# Where to mutate and how
targeton.tsv
ref_chr      chr17
ref_strand   -
ref_start    43115634
ref_end      43115878
r2_start     43115726
r2_end       43115779
ext_vector    "25,25"
action_vector "(),(aa,inframe,stop),(,)"

# The transcript model, 51 records
ENST00000357654.9.gtf
...
CDS 43115726 43115779 . - 1
...

# The reference genome
chr17.fa 81 MB

# Adaptors, strand, and annotation as flags
$ valiant sge targeton.tsv ../ref/chr17.fa out \
  'homo sapiens' 'GRCh38' \
  --revcomp-minus-strand \
  --adaptor-5 ... --adaptor-3 ... \
  --gff ENST00000357654.9.gtf

Inputs: 3 files
Outputs: oligos, genomic coordinates,
consequence annotation, and a VCF
```